Bardiya Aryanfard

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RESEARCH INTERESTS

Graph Algorithms, Differential Privacy

EDUCATION

INSTITUTE OF SCIENCE AND TECHNOLOGY AUSTRIA | PhD Candidate in Computer Science
Klosterneuburg, Austria | Sep 2023 – Present
Supervisor: Monika Henzinger | Research on differential privacy and graph algorithms, including theoretical analysis and empirical evaluation of private learning algorithms.

SHARIF UNIVERSITY OF TECHNOLOGY | B.Sc. in Computer Science
Tehran, Iran | Sep 2018 – Feb 2023
GPA: 18.83/20

WORK EXPERIENCE

PINKET | Front-end Web Developer
Tehran, Iran | Apr 2020 – Sep 2020

DAAL NAVIGATOR | Research and Development Intern
Tehran, Iran | Jul 2019 – Nov 2019

RESEARCH EXPERIENCE

IIT HYDERABAD | Research Assistant to Dr. Fahad Panolan
Jul 2021 – Jan 2022
Worked on parameterized complexity of problems related to coloring.

IIT GANDHINAGAR | Research Assistant to Dr. Manoj Gupta
Feb 2021 – Jul 2021
Worked on shortest path algorithms in fault-tolerant graphs.

PUBLICATIONS

- Connectivity in a New Bounded-Space Dynamic Model
  Bardiya Aryanfard, Monika Henzinger, John Kuszmaul, Jakob Nogler, Virginia Vassilevska Williams
  Under submission. (Link to preprint)
- On Differentially Private Mechanisms for Linear Regression
  Bardiya Aryanfard, Monika Henzinger, Farhood Rostamkhani
  NeurIPS 2026. (Link to paper)
- Improved Lower Bounds for Privacy under Continual Release
  Bardiya Aryanfard, Monika Henzinger, David Saulpic, A. R. Sricharan
  PODS 2026. (Link to paper)
- Parameterized Complexity of List Coloring and Max Coloring
  Bardiya Aryanfard, Fahad Panolan
  CSR 2022. (Link to paper)

COURSE PROJECTS

Implementation of Heuristic Methods for the Travelling Salesman Problem
As part of the course *Introduction to Combinatorial Optimization*. GitHub Link

HONORS & AWARDS

- Ranked 1st in Iran's National Graduate University Entrance Examination for Computer Science (2023)
- Bronze Medal in 27th National Informatics Olympiad (2017)

SKILLS

- Programming: Python, Java, C/C++
- Tools: MATLAB, LaTeX, Ipe

TEACHING ASSISTANTSHIPS

- Introduction to Cryptography
- Data Structures and Algorithms
- Data Transportation and Networks

RELEVANT COURSEWORK

Grades are reported in the original university scale (Iran: 20/20 = highest; Austria: 1/5 = highest).

- Advanced Graph Algorithms (In progress)
- Algorithms for Data Science (1/5)
- Algorithms with Differential Privacy (Pass)
- Data Structures (20/20)
- Analysis of Algorithms (19.9/20)
- Distributed Systems (19.7/20)
- Introduction to Cryptography (18.7/20)
- Introduction to Combinatorial Optimization (19.2/20)
- Coding Theory (M.Sc.) (19.0/20)
- Approximation Algorithms (M.Sc.) (19.2/20)